

NAME: Raghavan.S. STD.: _____ SEC.: _____ ROLL NO.: 520 SUB.: MAD-LAB

S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
1.		Kotlin Basics	2	✓
2.		Functions	4	
3.		Classes and object	6	
4.		Build first Android App	8	✓
5.		Layouts	10	
6.		App Navigation	12	✓
7.		Advanced RecyclerViews core	14	
8.		Connect to the Internet	16	✓
9.		Repository Pattern and ^{work} design	18	
10.		APP UI Design	20	

Completed

Mr. [Signature]

LAB 1

Build First Android App

Enter first text

aabc

Enter second text

aabc

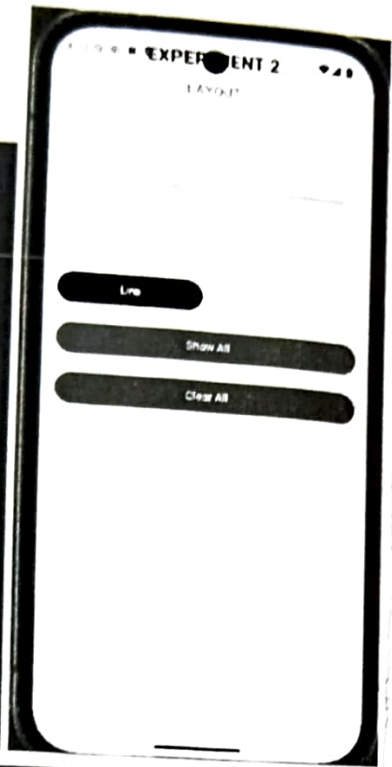
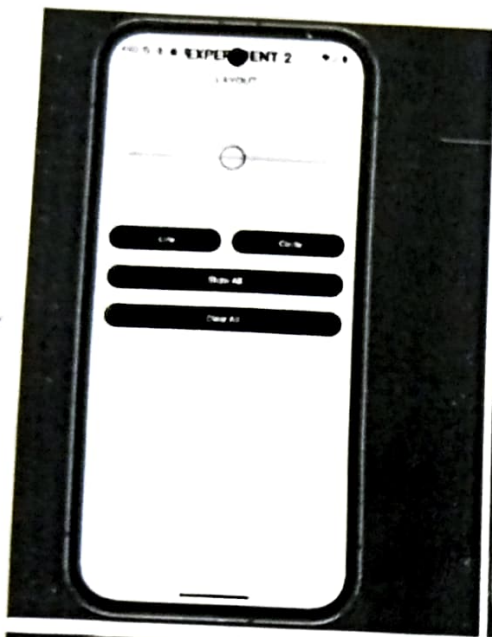
Date :

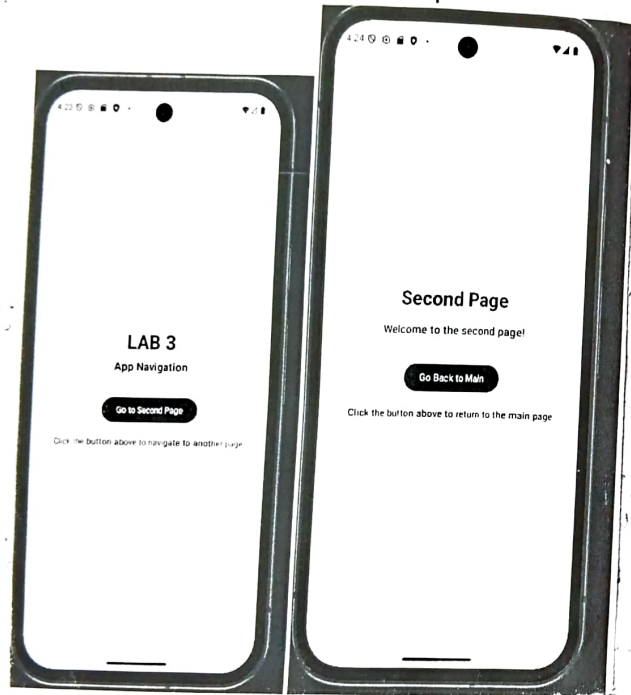
Aim :

To understand the basic Syntax and features of Kotlin Programming language.

Algorithm :

1. Start the Android Application.
2. Create a new Android project in Android Studio using Kotlin and Jetpack compose.
3. Import the required Packages and libraries.
4. Run the Application in a Android





classes and object

Ex. no: 3

Date :

Aim : To understand object-oriented programming concepts using classes and object in Kotlin.

Algorithm :

1. Start the Android Application.
2. Create new Android project in Android Studio using Kotlin and Jetpack Compose.
3. Import the required Packages and libraries.
4. Run the application in an Android emulator or Physical device.
5. Verify the output.
6. Stop the Program.

Dr. H.

Build first Android App

Ex. no: 14

Date: .

Aim:

To create and run a Simple Android Application using Android Studio and Kotlin

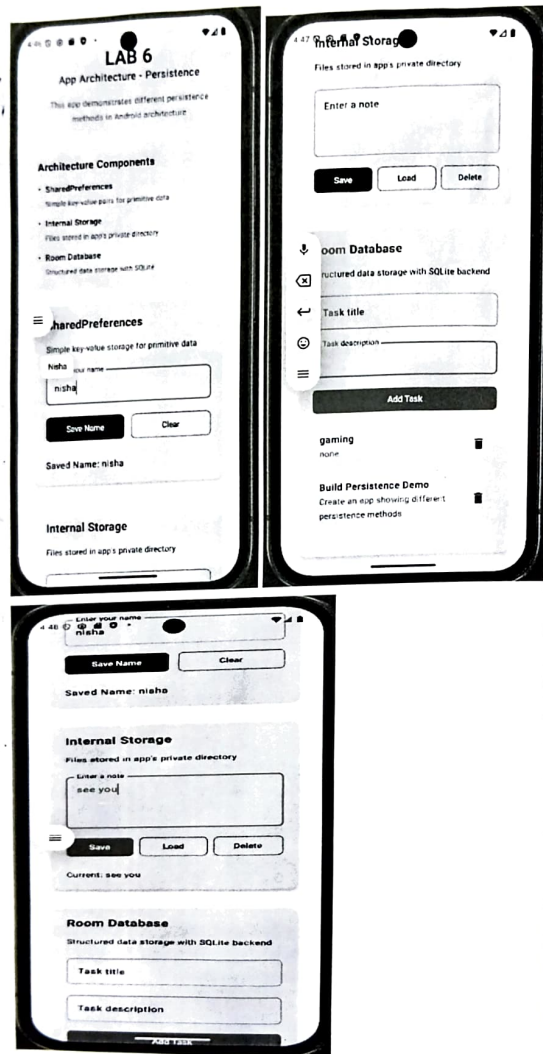
Algorithm:

1. Start the Android Application.
2. Create a new Android project in Android Studio using Kotlin and Jetpack Compose.
3. Import the required Packages and libraries.
4. Run the application in an Android emulator or Physical device.
5. Verify the output.
6. Stop the Program.

Result:

A basic Android application was successfully created and executed.





Ex.no:6

App Navigation

Date:

Aim:

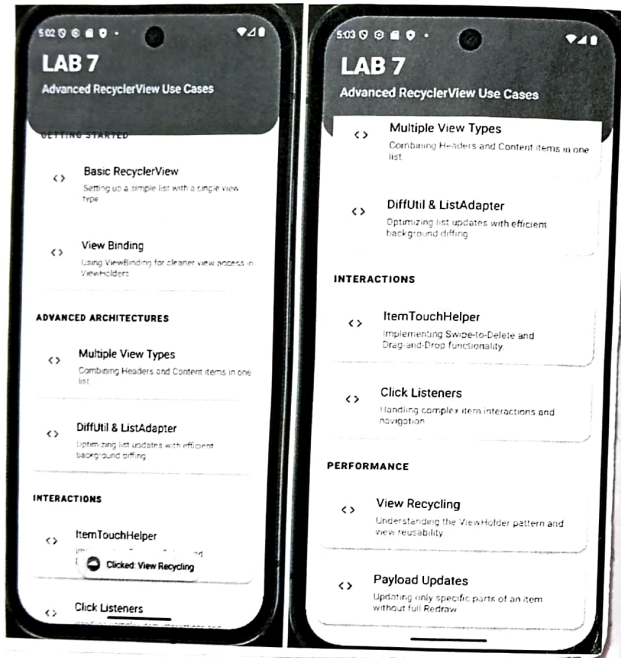
To implement navigation between different screen in an Android application

Algorithm:

1. Start the Android Application
2. Create a new Android Project in Android Studio using Kotlin and Jetpack Compose.
3. Import the required packages and libraries.
4. Run the application in an Android emulator or physical device.
5. Verify the output.
6. Stop the Program.

Result:

Navigation between multiple screen was implemented successfully.



Exno: 7

Date:

Assignment

Advanced RecyclerView use cases.

Aim:

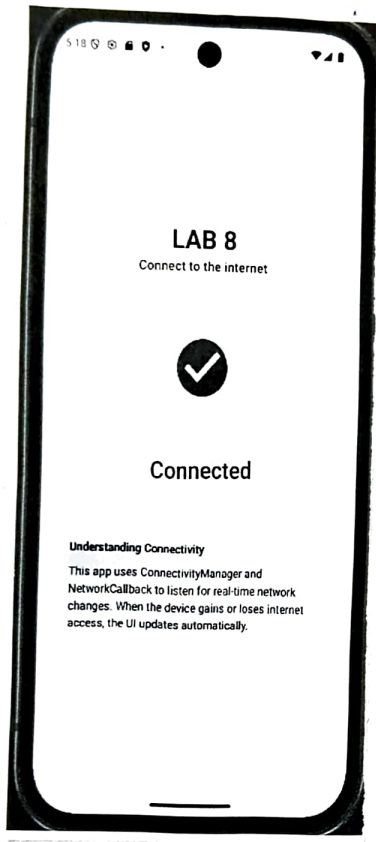
To study the Advanced RecyclerView use cases in Android

Algorithm:

1. Start the Android application and open Android Studio to create a Kotlin Project.
2. Add the RecyclerView dependency and design a layout containing a RecyclerView.
3. Create a data model and an adapter with ViewHolder to display the list items.
4. Add item interactions such as click listeners and swipe actions using item helpers.
5. Run the application and verify output.

Result:

Advanced RecyclerView use cases were implemented and was observed and understood successfully.



Ex no. 8

Connect to the Internet

Date :

Aim :

To connect the Android application to the internet and retrieve data from online sources.

Algorithm :

- * To connect the Android application to the internet and retrieve.
- * Start the Android Application and create a new project in Android Studio using Kotlin.
- * Add the required internet and network permission in the file.
- * Use ConnectivityManager to monitor the network.
- * Implement network callback to detect when the device connects on internet.
- * Update the UI to display the connectivity status.
- * Run the application and verify the states.

Result :

The application successfully connected to the Internet.



Expt 2 Repository Pattern and Background Worker

Aim:

To implement the Repository Pattern and Background Worker for background in Android

Algorithm:

- * Start the Android application and create a new Project in Android Studio using Kotlin
- * Create a Room database and implement the Repository Pattern to manage data operations.
- * Connect the Repository with worker to store update data in database.
- * Run the application and verify.

Result:

Repository Pattern and worker were implemented successfully for data management and background Process.

